

Docker telepítés és különböző funkciók

Docker telepítés, tesztelések:

<https://github.com/viraghkrisztian-pataky/Linux-vizsga/blob/main/docker/READ.ME?authuser=0>

```
apt install ca-certificates curl gnupg lsb-release gpg -y
```

```
mkdir -p /etc/apt/keyrings
```

```
curl -fsSL https://download.docker.com/linux/debian/gpg | gpg --dearmor -o  
/etc/apt/keyrings/docker.gpg
```

```
echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg]  
https://download.docker.com/linux/debian $(lsb_release -cs) stable" | tee  
/etc/apt/sources.list.d/docker.list > /dev/null
```

```
apt update
```

```
apt install -y docker-ce docker-ce-cli containerd.io docker-compose-plugin
```

```
systemctl enable docker
```

```
systemctl start docker
```

```
systemctl status docker
```

```
root@debian:~# systemctl enable docker
Synchronizing state of docker.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable docker
root@debian:~# systemctl start docker
root@debian:~# systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-02-12 10:51:07 CET; 1min 58s ago
     TriggeredBy: ● docker.socket
    Docs: https://docs.docker.com
   Main PID: 16493 (dockerd)
    Tasks: 10
   Memory: 36.1M
     CPU: 1.195s
   CGroup: /system.slice/docker.service
           └─16493 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Feb 12 10:51:06 debian dockerd[16493]: time="2026-02-12T10:51:06.025387953+01:00" level=info msg="Restoring container"
Feb 12 10:51:06 debian dockerd[16493]: time="2026-02-12T10:51:06.142211750+01:00" level=info msg="Deleting nftables"
Feb 12 10:51:06 debian dockerd[16493]: time="2026-02-12T10:51:06.177857544+01:00" level=info msg="Deleting nftables"
Feb 12 10:51:07 debian dockerd[16493]: time="2026-02-12T10:51:07.089899659+01:00" level=info msg="Loading containers"
Feb 12 10:51:07 debian dockerd[16493]: time="2026-02-12T10:51:07.107533429+01:00" level=info msg="Docker daemon" com
Feb 12 10:51:07 debian dockerd[16493]: time="2026-02-12T10:51:07.107850557+01:00" level=info msg="Initializing build"
Feb 12 10:51:07 debian dockerd[16493]: time="2026-02-12T10:51:07.151362115+01:00" level=info msg="Completed buildkit"
Feb 12 10:51:07 debian dockerd[16493]: time="2026-02-12T10:51:07.168131424+01:00" level=info msg="Daemon has complet
Feb 12 10:51:07 debian dockerd[16493]: time="2026-02-12T10:51:07.168864238+01:00" level=info msg="API listen on /run
Feb 12 10:51:07 debian systemd[1]: Started docker.service - Docker Application Container Engine.
lines 1-22/22 (END)
```

Felhasználó hozzáadása

usermod -aG docker martin

Az első teszt docker futtatása

docker run hello-world

kiírja a válasz üzenetet

```
root@debian:~# docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
ea52d2000f90: Download complete
17eec7bbc9d7: Pull complete
Digest: sha256:ef54e839ef541993b4e87f25e752f7cf4238fa55f017957c2eb44077083d7a6a
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

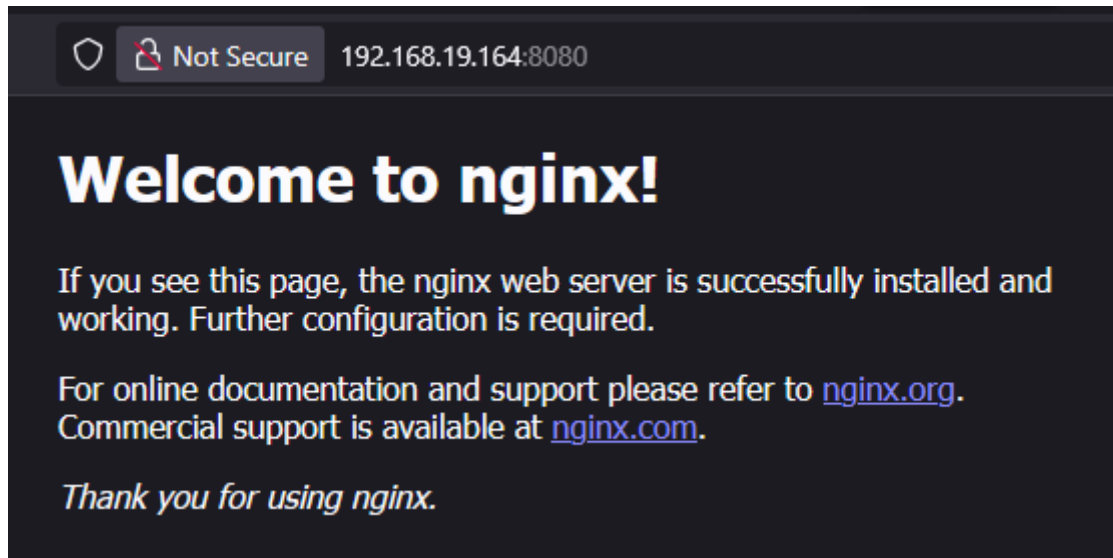
For more examples and ideas, visit:
https://docs.docker.com/get-started/

root@debian:~# █
```

docker run -d -p 8080:80 nginx

docker ps -a

```
root@debian:~# docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
f3c1e248ee3a   nginx     "/docker-entrypoint..." 38 seconds ago Up 37 seconds 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp nifty_hellman
adf59a50ef77   hello-world "/hello"                About a minute ago Exited (0) About a minute ago exciting_ishizaka
root@debian:~#
```



ez törlés csak próba volt

docker stop ID

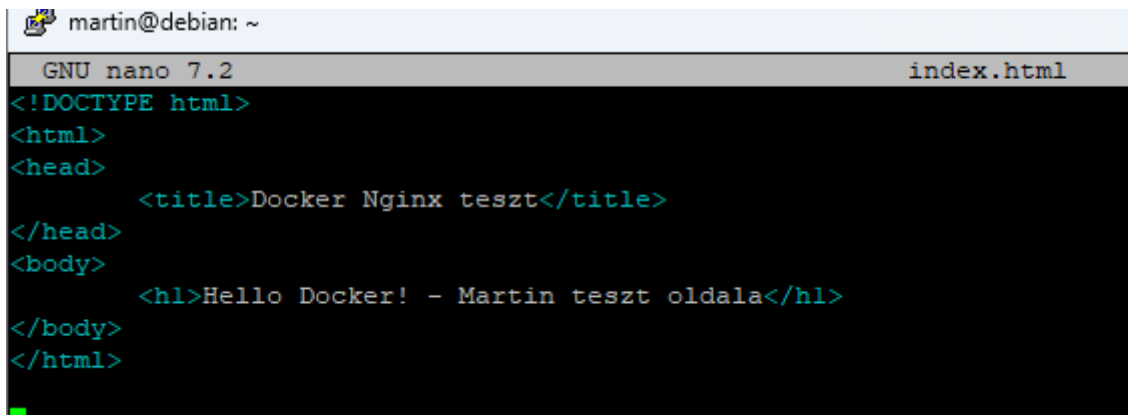
docker rm ID

1. NGNIX weboldal

Mappa létrehozása:

```
mkdir nginx-static
```

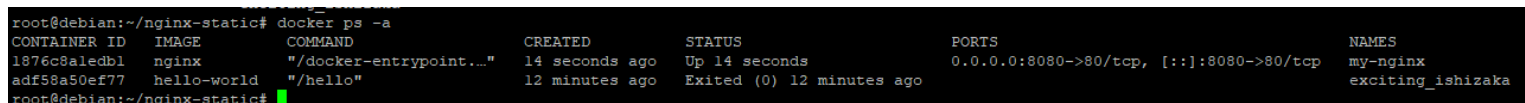
```
cd nginx-static
```



```
martin@debian: ~  
GNU nano 7.2 index.html  
<!DOCTYPE html>  
<html>  
<head>  
  <title>Docker Nginx teszt</title>  
</head>  
<body>  
  <h1>Hello Docker! - Martin teszt oldala</h1>  
</body>  
</html>
```

```
<!DOCTYPE html>  
<html>  
  <head>  
    <title>Docker Nginx teszt</title>  
  </head>  
  <body>  
    <h1>Hello Docker! - Martin teszt oldala</h1>  
  </body>  
</html>
```

```
docker run -d -p 8080:80 -v $(pwd):/usr/share/nginx/html --name my-nginx nginx
```



```
root@debian:~/nginx-static# docker ps -a  
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES  
l876c8aledb1   nginx    "/docker-entrypoint..." 14 seconds ago Up 14 seconds 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp my-nginx  
adf58a50ef77   hello-world "/hello"                12 minutes ago Exited (0) 12 minutes ago exciting_ishizaka  
root@debian:~/nginx-static#
```



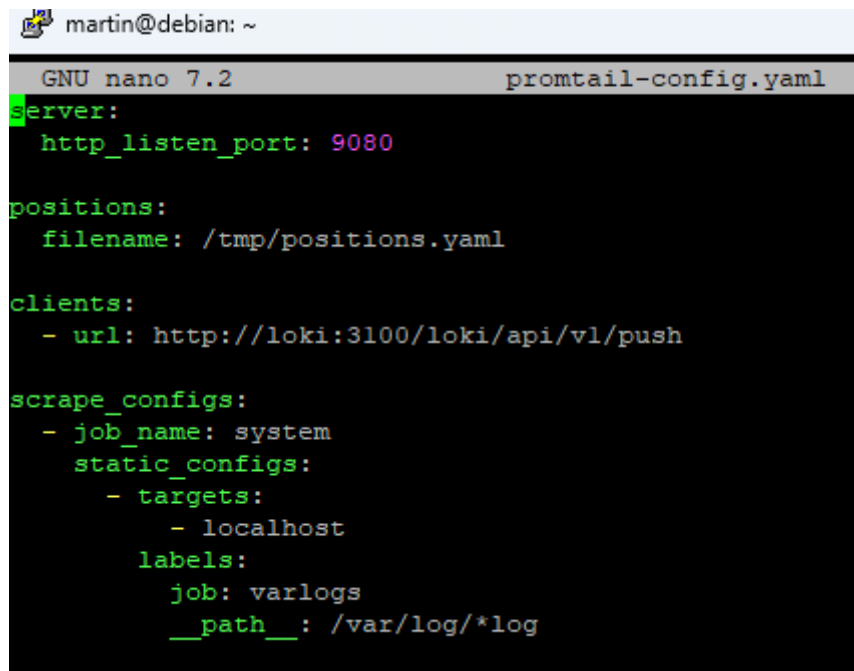
2. Grafana + Loki

Mappa létrehozása:

```
mkdir grafana-loki
```

```
cd grafana-loki
```

Első fájl: `docker-compose.yml`



```
martin@debian: ~  
GNU nano 7.2 promtail-config.yaml  
server:  
  http_listen_port: 9080  
  
positions:  
  filename: /tmp/positions.yaml  
  
clients:  
  - url: http://loki:3100/loki/api/v1/push  
  
scrape_configs:  
  - job_name: system  
    static_configs:  
      - targets:  
        - localhost  
        labels:  
          job: varlogs  
          __path__: /var/log/*log
```

version: "3.9"

services:

loki:

image: grafana/loki:2.9.0

ports:

- "3100:3100"

command: -config.file=/etc/loki/local-config.yaml

promtail:

image: grafana/promtail:2.9.0

volumes:

- /var/log:/var/log

- ./promtail-config.yaml:/etc/promtail/promtail-config.yaml

command: -config.file=/etc/promtail/promtail-config.yaml

grafana:

image: grafana/grafana:latest

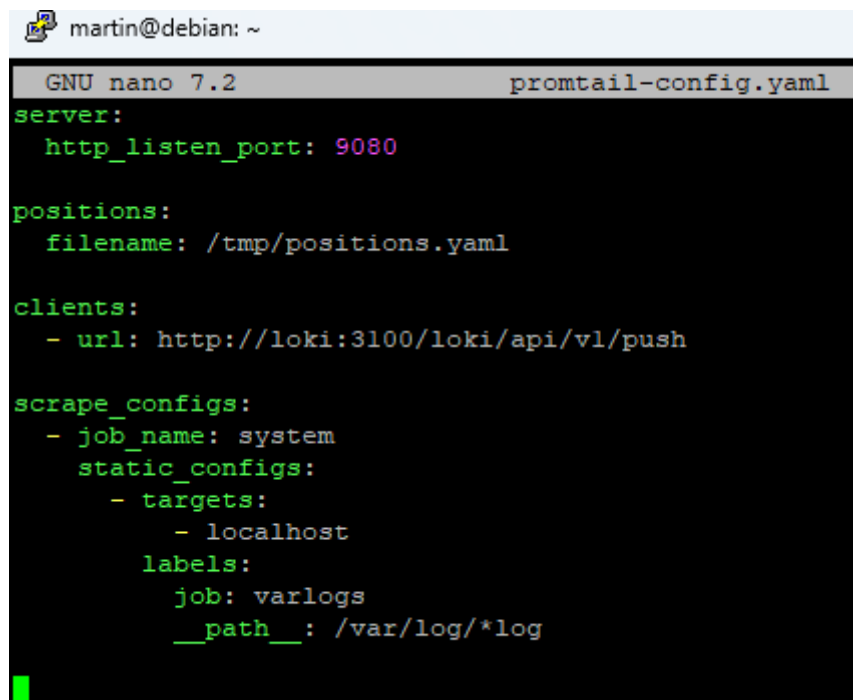
ports:

- "3000:3000"

depends_on:

- loki

Második fájl: promtail-config.yaml



```
martin@debian: ~  
GNU nano 7.2 promtail-config.yaml  
server:  
  http_listen_port: 9080  
  
positions:  
  filename: /tmp/positions.yaml  
  
clients:  
  - url: http://loki:3100/loki/api/v1/push  
  
scrape_configs:  
  - job_name: system  
    static_configs:  
      - targets:  
        - localhost  
        labels:  
          job: varlogs  
          __path__: /var/log/*log
```

server:

http_listen_port: 9080

positions:

filename: /tmp/positions.yaml

clients:

- url: http://loki:3100/loki/api/v1/push

scrape_configs:

- job_name: system

static_configs:

- targets:

- localhost

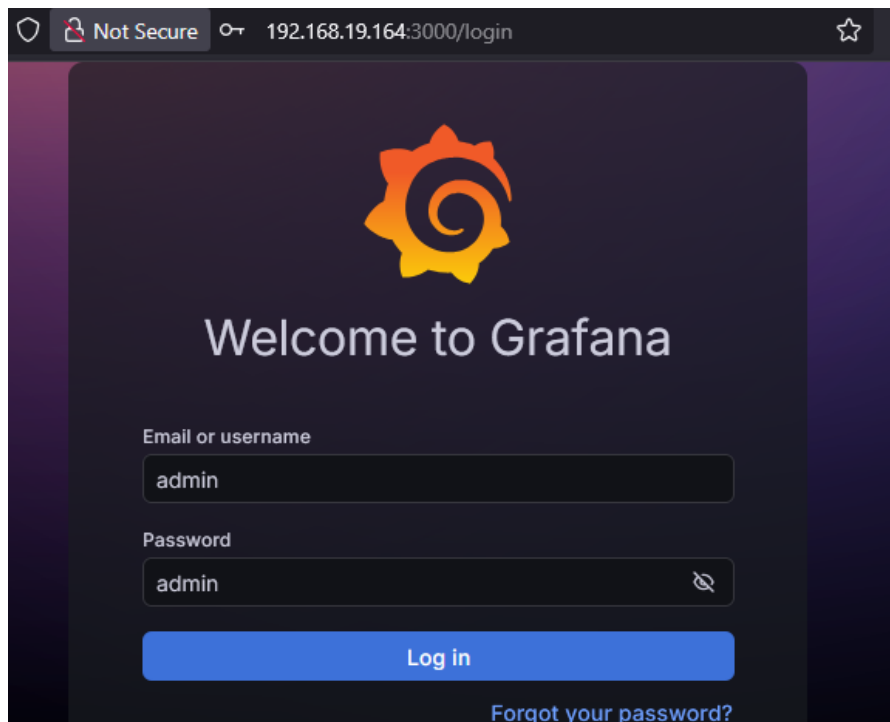
labels:

job: varlogs

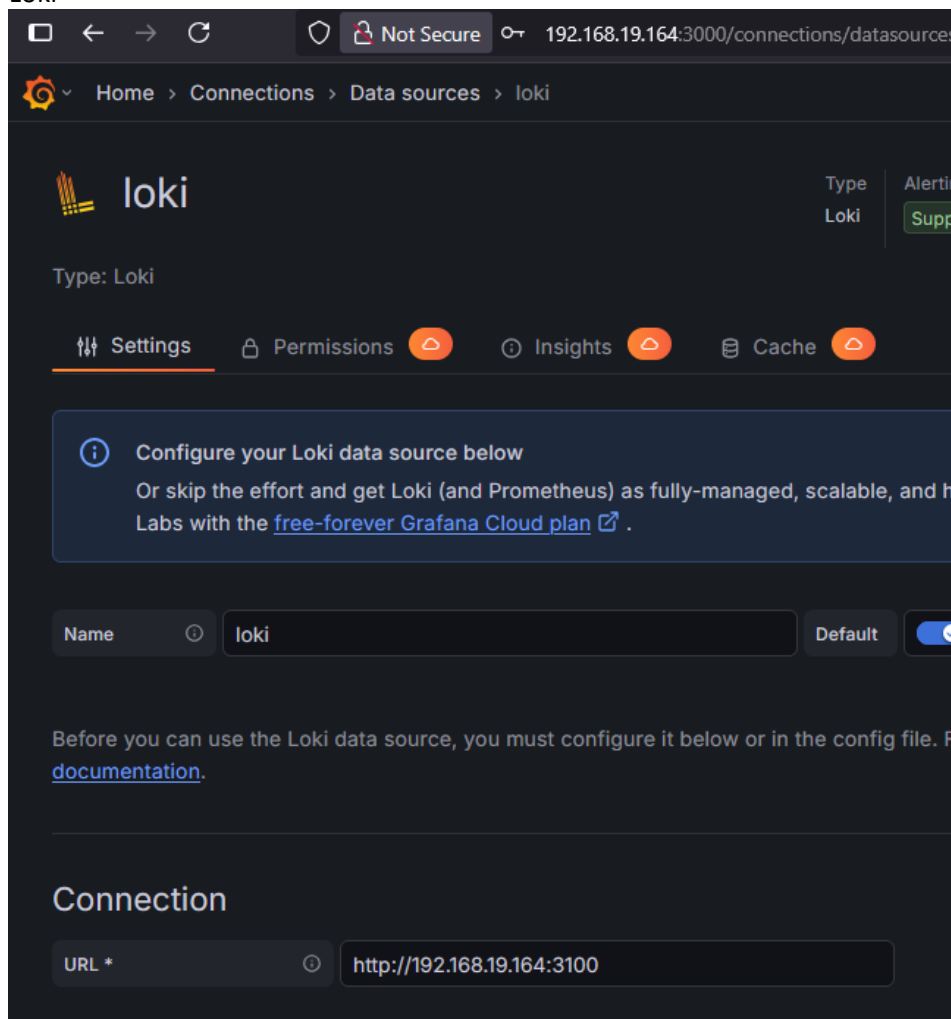
__path__: /var/log/*log

Futtatás:

docker compose up -d

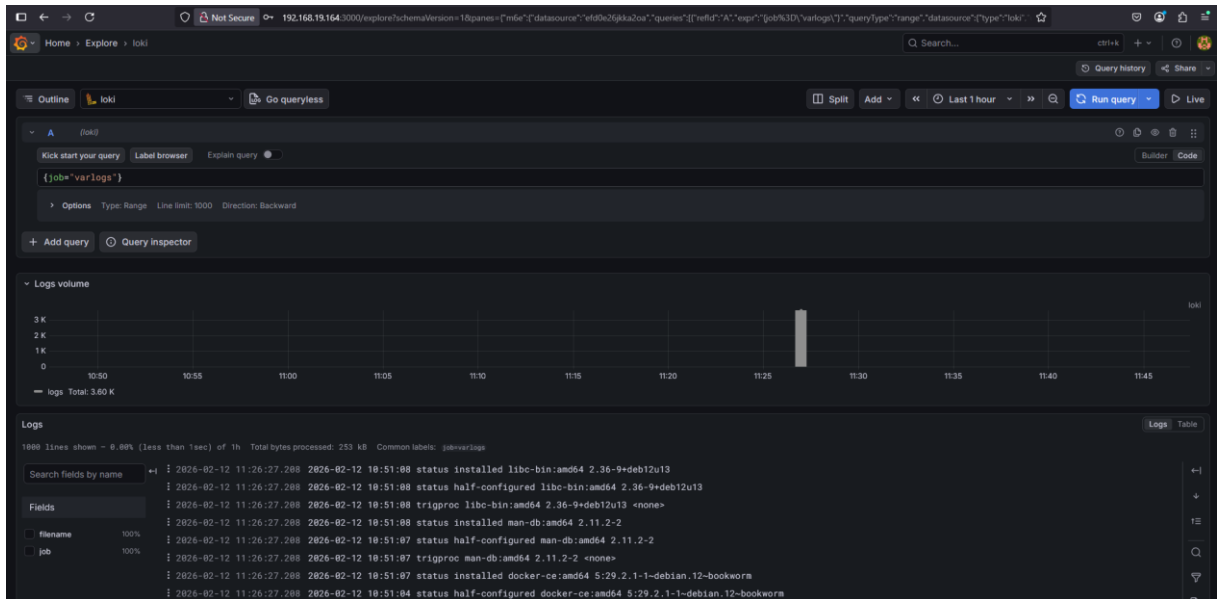


Loki



save

Loki üzemel



3. Portainer telepítése:

Docker volume létrehozása Portainer számára

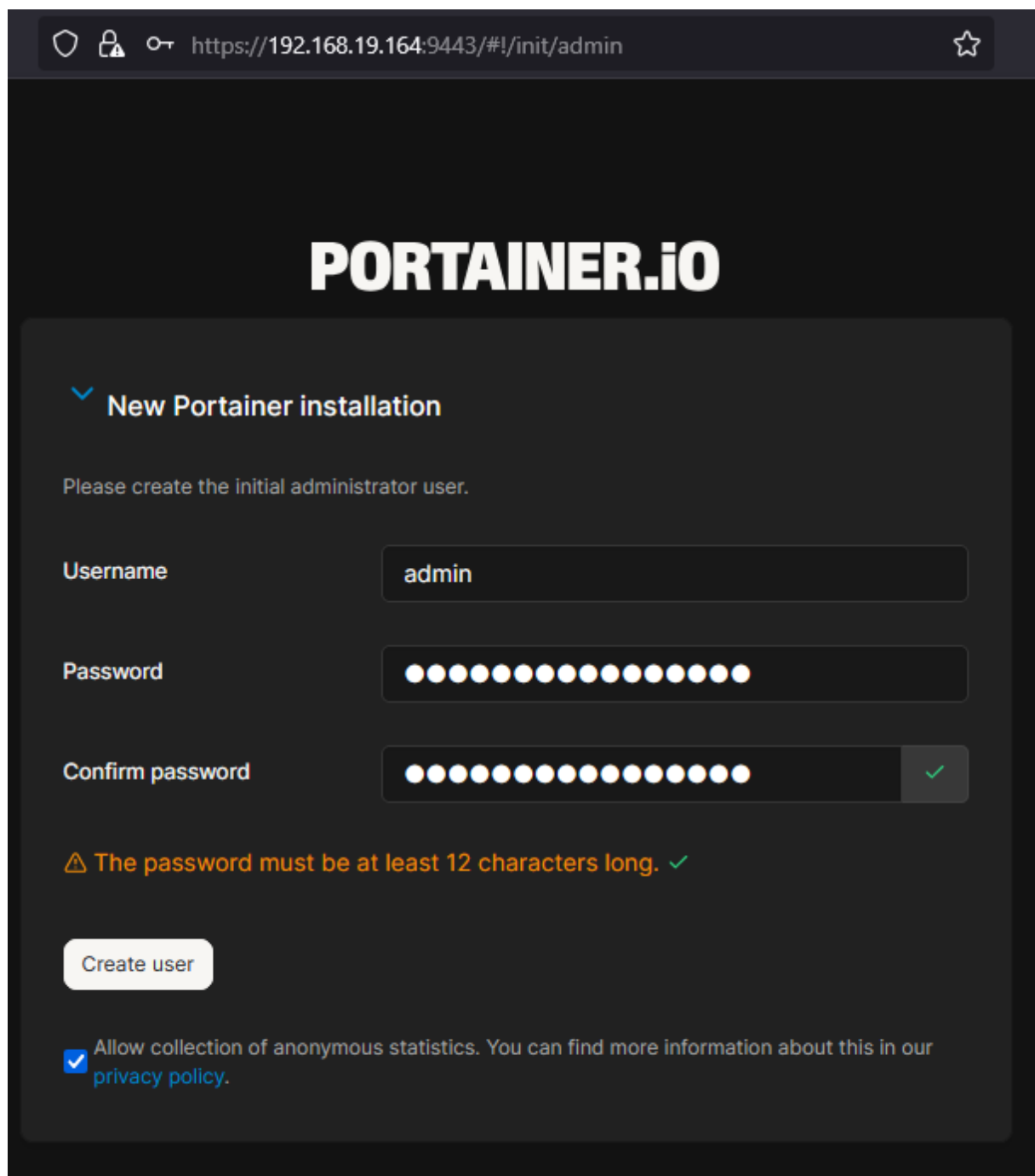
```
docker volume create portainer_data
```

Portainer telepítése

```
docker run -d \  
-p 9443:9443 \  
-p 8000:8000 \  
--name portainer \  
--restart=always \  
-v /var/run/docker.sock:/var/run/docker.sock \  
-v portainer_data:/data \  
portainer/portainer-ce:latest
```

```
martin@debian: ~  
root@debian:~# docker volume create portainer_data  
portainer_data  
root@debian:~# docker run -d \  
-p 9443:9443 \  
-p 8000:8000 \  
--name portainer \  
--restart=always \  
-v /var/run/docker.sock:/var/run/docker.sock \  
-v portainer_data:/data \  
portainer/portainer-ce:latest  
Unable to find image 'portainer/portainer-ce:latest' locally  
latest: Pulling from portainer/portainer-ce  
4f4fb700ef54: Pull complete  
c2b6c8eed7f1: Pull complete  
f87513245ccf: Pull complete  
7999c6081b17: Pull complete  
3ac858128074: Pull complete  
5facec6e6233: Pull complete  
3129b44812ef: Pull complete  
5a84dd07f347: Pull complete  
f793926a441c: Download complete  
29fcb1a18bf8: Download complete  
Digest: sha256:9012a4256c4632f2c6162da361a4d4db9d6d04800e0db0137de96e31656ab876  
Status: Downloaded newer image for portainer/portainer-ce:latest  
eb3727e66dfd66b0b604109a27ce6de319be5a133a336bcd1bc9fa4faadaec7e  
root@debian:~#
```

https !!!!!



The screenshot shows a web browser window with the address bar displaying `https://192.168.19.164:9443/#!/init/admin`. The main heading is **PORTAINER.io**. Below it, a section titled **New Portainer installation** (indicated by a blue checkmark icon) contains the instruction: "Please create the initial administrator user." The form includes three input fields: "Username" with the value "admin", "Password" with 12 masked dots, and "Confirm password" with 12 masked dots and a green checkmark icon on the right. Below the fields, a warning message in orange text states: "⚠ The password must be at least 12 characters long. ✓". At the bottom left is a "Create user" button. At the bottom right is a checkbox that is checked, with the text "Allow collection of anonymous statistics. You can find more information about this in our [privacy policy](#)."

https://192.168.19.164:9443/#!/init/admin

PORTAINER.io

✓ New Portainer installation

Please create the initial administrator user.

Username

Password

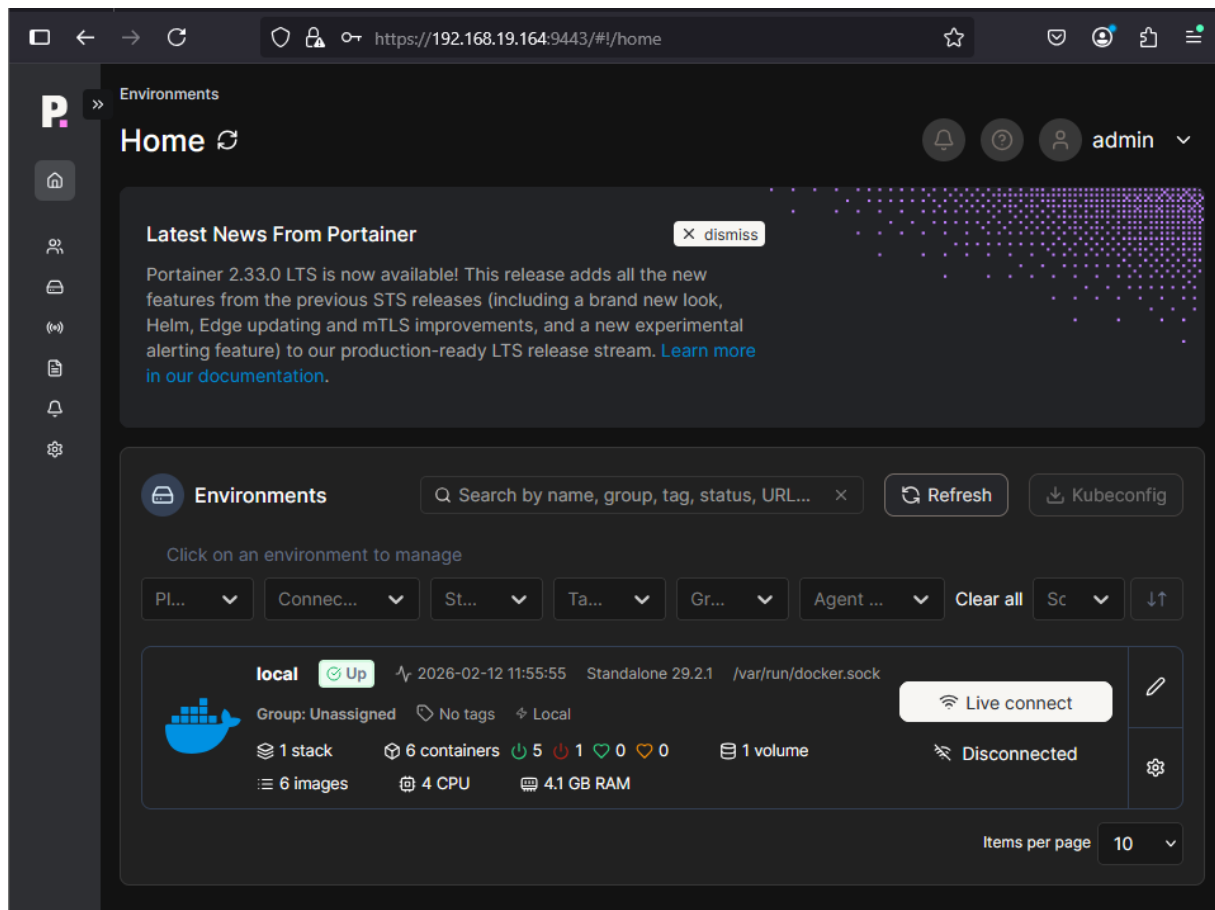
Confirm password ✓

⚠ The password must be at least 12 characters long. ✓

Create user

☒ Allow collection of anonymous statistics. You can find more information about this in our [privacy policy](#).

admin – Jelszo12Jelszo12



Futó konténereink:

